

3.4 BANDWIDTH CONTROLS OPTIMIZATION ACCESS TO HIGH PRECISION

The construction’s nearly width-independent relative bandwidth $\lambda = \gamma h$ requires $\gamma = \Theta(W)$. Even though larger widths admit smaller λ at comparable construction error (Figure 2b), small slopes can delay readout optimization. We quantify this spectral-bias mechanism (Xu et al., 2020; Tancik et al., 2020) in the dictionary of Section 3.1, replacing the direct normalized least-squares solve with GD while keeping the features fixed.

Theorem 3.2 (Slope-dependent output-error lower bound). *For the fixed-center construction of Section 3.1 with common slope $\gamma > 0$, let $\mu_1 \geq \mu_2 \geq \dots \geq 0$ be the normalized readout-kernel eigenvalues. For a nonzero sampled target, readout GD from zero with $\eta = 1/(2\mu_1)$ satisfies*

$$\frac{\|f - q_n\|_{2,Z}^2}{\|f\|_{2,Z}^2} \geq \sum_i p_i(\gamma) \left(1 - \frac{\bar{\rho}_i(\gamma)}{2}\right)^{2n}, \quad n \geq 0. \quad (1)$$

Here $\|\cdot\|_{2,Z}$ is the training-sample RMS norm, p_i is the fraction of sampled target energy projected onto kernel eigenvector i , and $\mu_i/\mu_1 \leq \bar{\rho}_i(\gamma) \leq 1$. The explicit gamma-dependent rate bounds and finite-geometry allowances are given in Appendix A.

Proof sketch. The readout kernel contains normalized inner products of feature vectors at pairs of training points. After removing the constant sample direction, its center-integral approximation depends explicitly on $d \coth(\gamma d)$, where d separates those points. Bounding the finite-center corrections gives $\bar{\rho}_i(\gamma)$; standard GD eigenmode decay then yields the output-error bound.

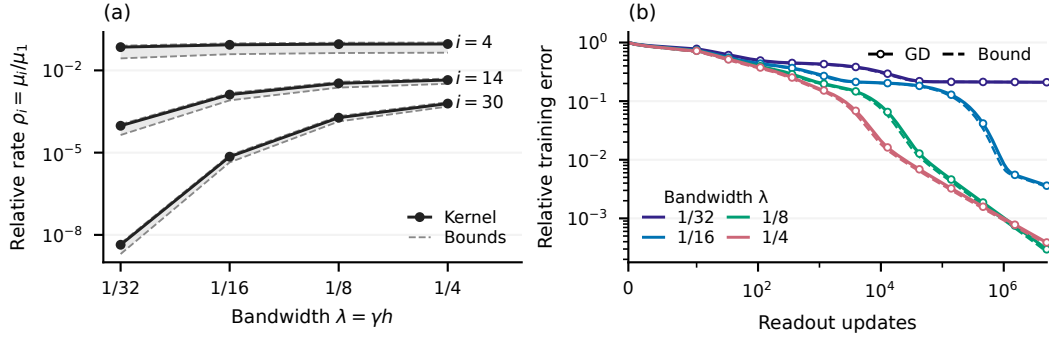


Figure 3: **Small slopes weaken kernel directions and delay readout learning.** Width $W = 512$; target $f(x) = [\sin(2\pi x) + \frac{1}{2}\sin(6\pi x) + \frac{1}{4}\sin(14\pi x)]/\sqrt{21/32}$ on $[-1, 1]$. **(a)** Kernel ratios (dots) and theorem intervals (dashed boundaries, shading) for three ranks carrying substantial target energy. Lines connect ordered ranks; eigenvectors change with bandwidth. **(b)** Executed frozen-readout GD errors (solid, hollow markers) and square roots of the theorem’s lower bounds (dashed), using all resolved target projections. Errors are relative training RMS through five million updates; geometry and protocol are in Appendix B.

Figure 3(a) selects ranks 4, 14, 30 because together they carry 59–63% of sampled target energy across the four bandwidths, including at least 8% in rank 14 and 2.5% in rank 30. Increasing λ from $1/32$ to $1/4$ raises their relative rates by factors of 1.3, 47, and 1.4×10^5 , respectively.

At $\lambda = 1/32$, a direct readout fit attains 6.2×10^{-10} dense-grid relative error, yet five million GD updates retain training error 0.210 against the theorem’s lower bound 0.2098 (Figure 3(b)). At $\lambda = 1/4$, GD reaches 3.88×10^{-4} over the same budget. Across the four bandwidths, the bound remains at least 75% of executed error throughout training: small slopes delay accuracy already available in the dictionary.

3.5 TRAINING MUST ACQUIRE USEFUL SLOPES AND CENTERS

Joint training also leaves a substantial precision gap over this budget. After five million updates, joint Adam reaches median dense-grid relative error 1.81×10^{-3} , versus 6.62×10^{-7} on the supplied $\lambda = 1/4$ features (Figure 4A).

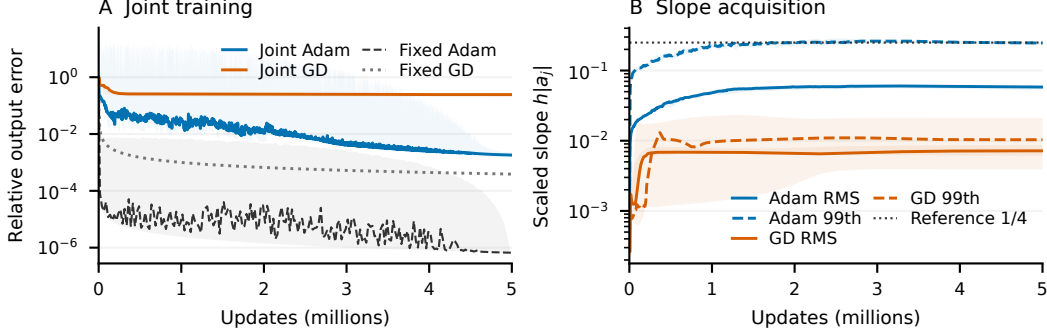


Figure 4: **Partial slope acquisition leaves a precision gap.** **A:** joint Adam/GD relative output errors and frozen uniform $\lambda = 1/4$ references under the same five-million-update budget. **B:** RMS and 99th-percentile scaled slopes $h|a_j|$ in A’s joint runs. The reference is a supplied geometry, not a necessary slope threshold. Joint curves summarize five paired seeds; shading retains seed variation and recorded within-bin extrema. Cosine schedules span the full horizon. These are empirical comparisons; Appendix B gives the protocol.

Adam’s 99th-percentile scaled slope approaches $1/4$, but only 4–7 of 512 features reach that reference across the five seeds (Figure 4B). Readout refits on the learned features barely improve their error (Appendix B). Longer training helps, but acquiring a few steep features does not recover the supplied geometry’s accuracy within this budget. This motivates the next analysis: training must acquire useful slopes and centers together. The construction supplies both the feature geometry and the readout directly.